

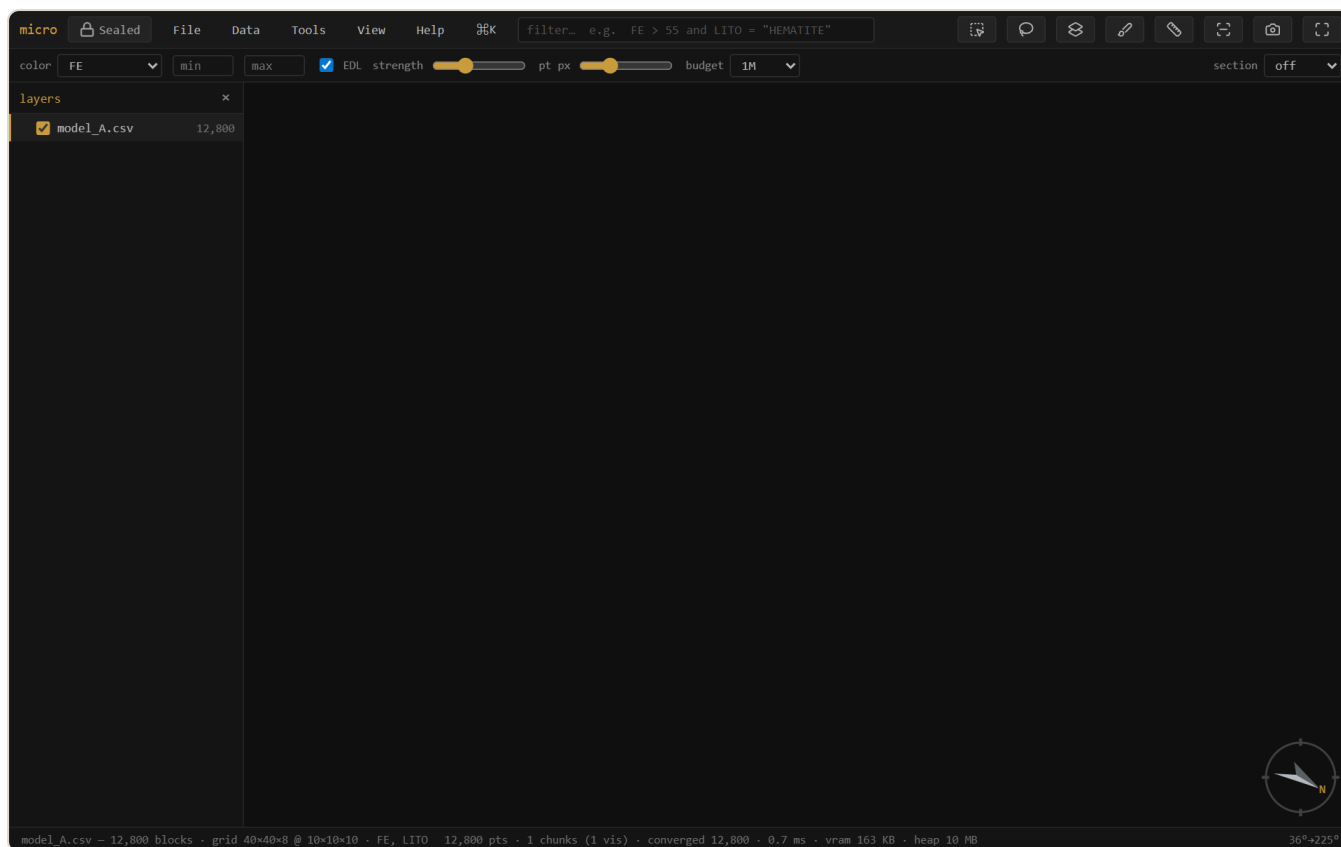
micro

A streaming, offline workbench for massive point clouds and block models.

 **Sealed** — no network, nothing leaves your machine · single self-contained HTML file · [download micro.html](#)
· [open the app](#) ↗

[User documentation](#) · [Auditable](#) / Geoscientific Chaos Union · gentropic.org/micro

micro opens a point cloud, block model, grid, or set of drillholes — however large — in about a second, with no import step, and lets you view, query, join, reconcile, validate, and produce report-ready figures from it. Every derived result carries a provenance trail, and nothing ever touches the network.



A 12,800-block iron model coloured by grade — the high-grade shoot in yellow. Loaded, inferred, and rendered from a plain CSV with no import step.

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1 What micro is

micro is a single self-contained HTML file — a **viewer that grew into a workbench**. It streams massive spatial-element files (never holding them fully in memory), so the first look is near-instant at any size and the picture densifies while you orbit.

It reads:

- **Point clouds** — LAS, PLY, XYZ / PTS / DAT dumps
- **Block models** — CSV / TXT, Datamine `.dm`, and **Parquet**
- **Grids** — GeoTIFF, Surfer `.grd`, ESRI ASCII `.asc`
- **Meshes** — `.msh` (Leapfrog), OBJ, PLY-with-faces, LFM
- **Drillholes** — collar + survey + interval tables (desurveyed on load)

Beyond viewing, it does the everyday resource-geology work: filter and section, domain by a solid, join and reconcile model versions, grade-tonnage and swath analysis, validation against drillholes, and figure export — all against the source file, all with a provenance trail, all offline.

Tip — The fastest way to reach anything is the **command palette**: `Ctrl/⌘K` (or the `⌘K` button). It searches every action and shows only what applies to the layer you have selected.

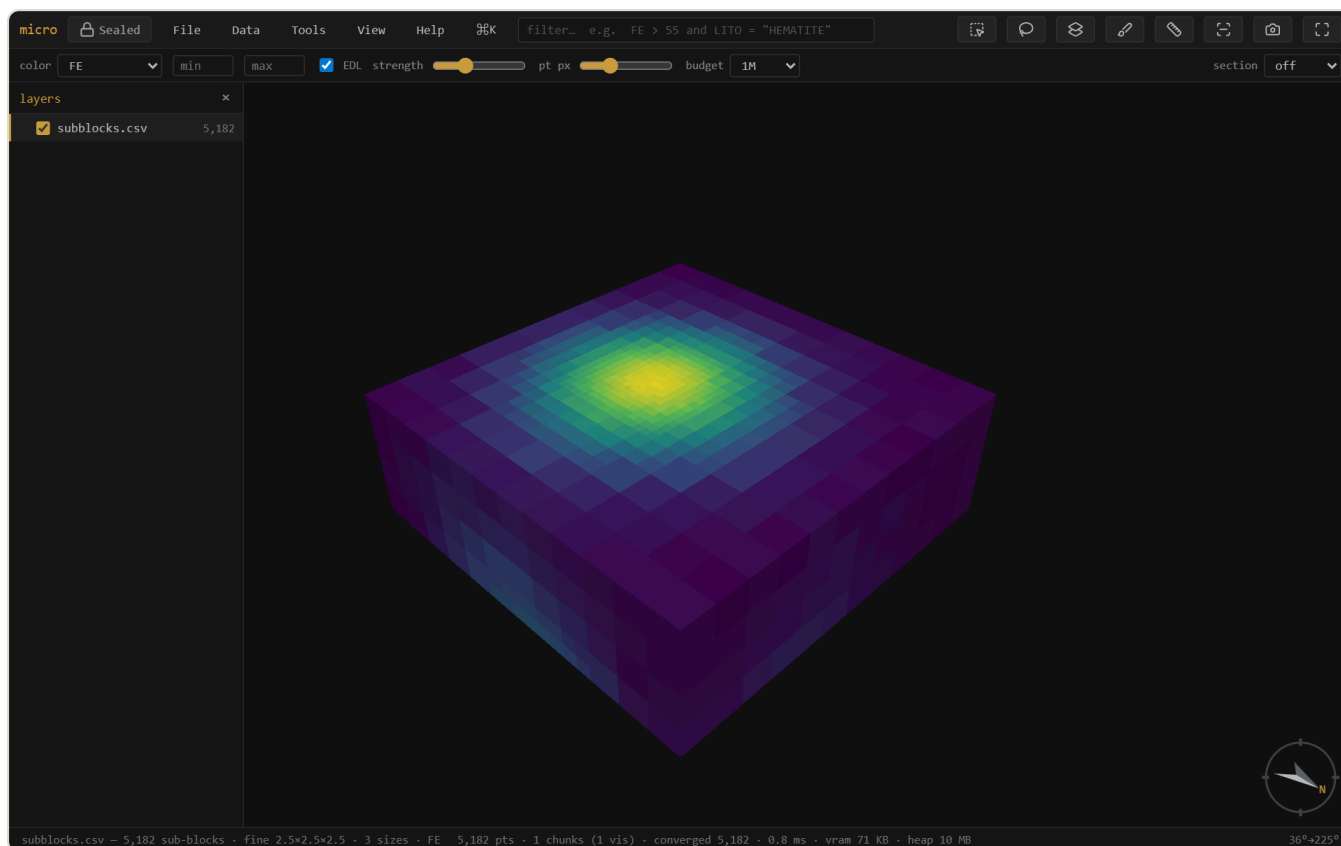
2 Getting started

Opening data

Drag a file (or several) anywhere onto the window, or use **File** → **Open....** Several files at once become separate **layers**. No data handy? **File** → **Sample bench scan** or **Sample block model** generate a dataset locally.

Sub-blocked models

micro reads **sub-blocked** (octree) block models — where cells are refined into smaller boxes near a boundary or a high-grade zone — and renders **each block at its true size**, straight from the file. CSV/Parquet models with per-block `dx/dy/dz` columns and Datamine `.dm` models with per-record `XINC/YINC/ZINC` are detected automatically; the model streams, picks, filters, and colours like any other. Join and reconcile handle them by **volume** — see Join & reconcile.



A sub-blocked (octree) model — coarse parent blocks away from the shoot, refined into progressively smaller boxes through the high-grade core, each drawn at its own size.

Note — Detection assumes an **axis-aligned** model with power-of-two (octree) refinement — the standard export. A **rotated** model currently opens as its centroids (points); interpreting a rotated model as boxes (enter or measure its azimuth) is on the roadmap.

The layers panel

p toggles the layers panel. Each opened dataset is a layer with its own colour, filter, and visibility; right-click a layer for its actions (zoom, info, attribute table, filter, analysis tools, remove). Layers can be grouped and reordered. **,** / **.** step the active layer.

Getting around

Drag to orbit, right-drag (or shift-drag) to pan, wheel to zoom. **f** fits the data. **n s e w** look level that way, **d** is plan, **o** toggles orthographic (needed for a true scale bar). Click a block or point to **pick** it — the record panel shows its full source row.

3 Viewing & querying

Colour

The **colour** dropdown drives the active layer: by elevation, a numeric **channel** (a grade), or a **category** (a rock type). Pick a ramp per layer (viridis · spectral · magma · turbo · greys) in **Properties** → **style**; the **min/max** boxes clip the scale so outliers stop eating the ramp. On a grade channel, double-click the histogram to add **breakpoints** — **classed** turns them into a solid-colour grade-shell legend.

Filter

focuses the filter box. It speaks a familiar expression language over every column: `FE > 55` and `LITO = "HEMATITE"`, `between`, `in (...)`. Matching elements isolate; everything else drops out of the render, the table, and any analysis. applies, clears.

Calculated columns & the scan

Properties → **columns** defines *f* **calculated columns** (`FE + 0.4 * MN`) that the filter, scan, table, and record panel all see, and **scan columns** streams the whole table once to fill `n · mean · std · quantiles · histogram` per column.

Sections & measure

A section is a live cut — everything hidden is neither drawn nor pickable. Pick a preset (plan · N–S · E–W), press and drag a line, or type a plane's dip-direction + dip. toggles it; step slabs. then two clicks **measures** 3D · plan · Δz between records.

The attribute table

opens the layer's rows as a draggable, layer-pinned window — windowed reads, so a 1.8M-row model scrolls cold. **matches** shows only the filtered rows; click a row to pick it in the scene.

4 Selection & domaining

Select with **r** (rectangle) or **v** (lasso): what you see gets selected (occlusion-honest, all visible layers), tinted gold — **shift** adds, **alt** subtracts, **Esc** clears. The **select-through** toolbar toggle flips to the extruded tube: everything whose position falls inside the shape, occluded included.

Select by solid / surface (Tools menu, or right-click a mesh) classifies records against a closed mesh by winding number: **select** inside/outside, write a **flag column**, or a **fraction-inside** 0–1 column. Open surfaces (topo, weathering horizons) classify **above / below / between**. **Assign domains** runs the whole coded-model loop — pick the domain solids and a column name, and every block goes to the solid holding its largest fraction.

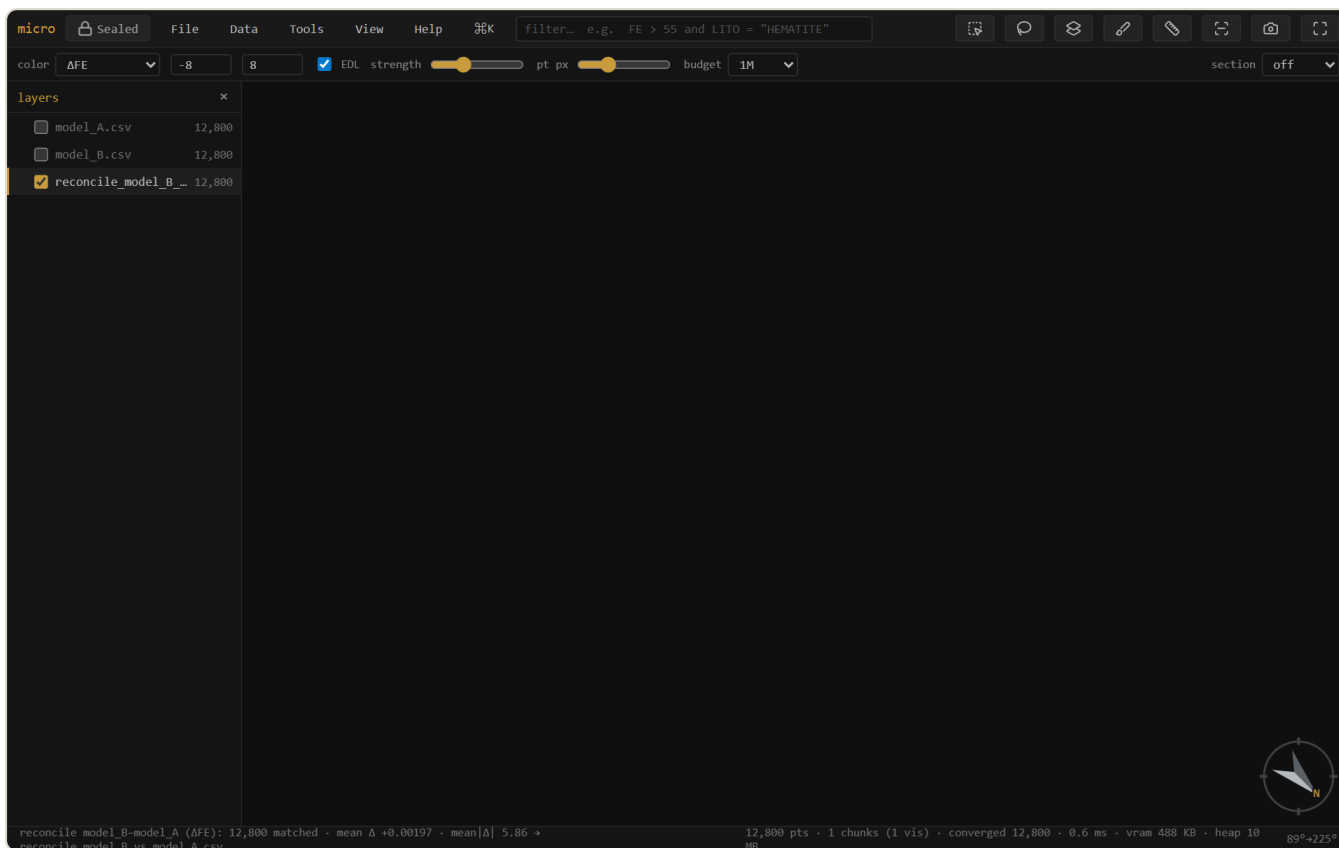
Paint a column (**b**) is interactive domaining by hand: a category column you fill by brushing over the scene. Painted values are real columns the filter, table, and export all see.

5 Join & reconcile

Two model versions? The **join tab** (Properties → **join**) compares them. A **spatial join** resamples a *compatible* model onto a target grid — a banner tells you whether the grids share a lattice (clean replicate / aggregate) or why they can't. A **key join** matches rows by a key column (e.g. a block ID, or a domain→density lookup).

Reconcile is the preset: it produces a coloured **Δ map** (this model – reference) on a common lattice, with a diverging ramp centred on zero — blue where the new estimate is lower, red where it's higher, neutral where they agree. The summary reads **matched · mean Δ · mean $|\Delta|$ · added · removed**.

Sub-blocked models reconcile by **volume**: each variable-size block is aggregated onto the reference (parent) grid weighted by the space it fills — a big parent and its small sub-blocks each contribute in proportion — so a sub-blocked estimate compares cleanly against a regular one at the parent resolution.

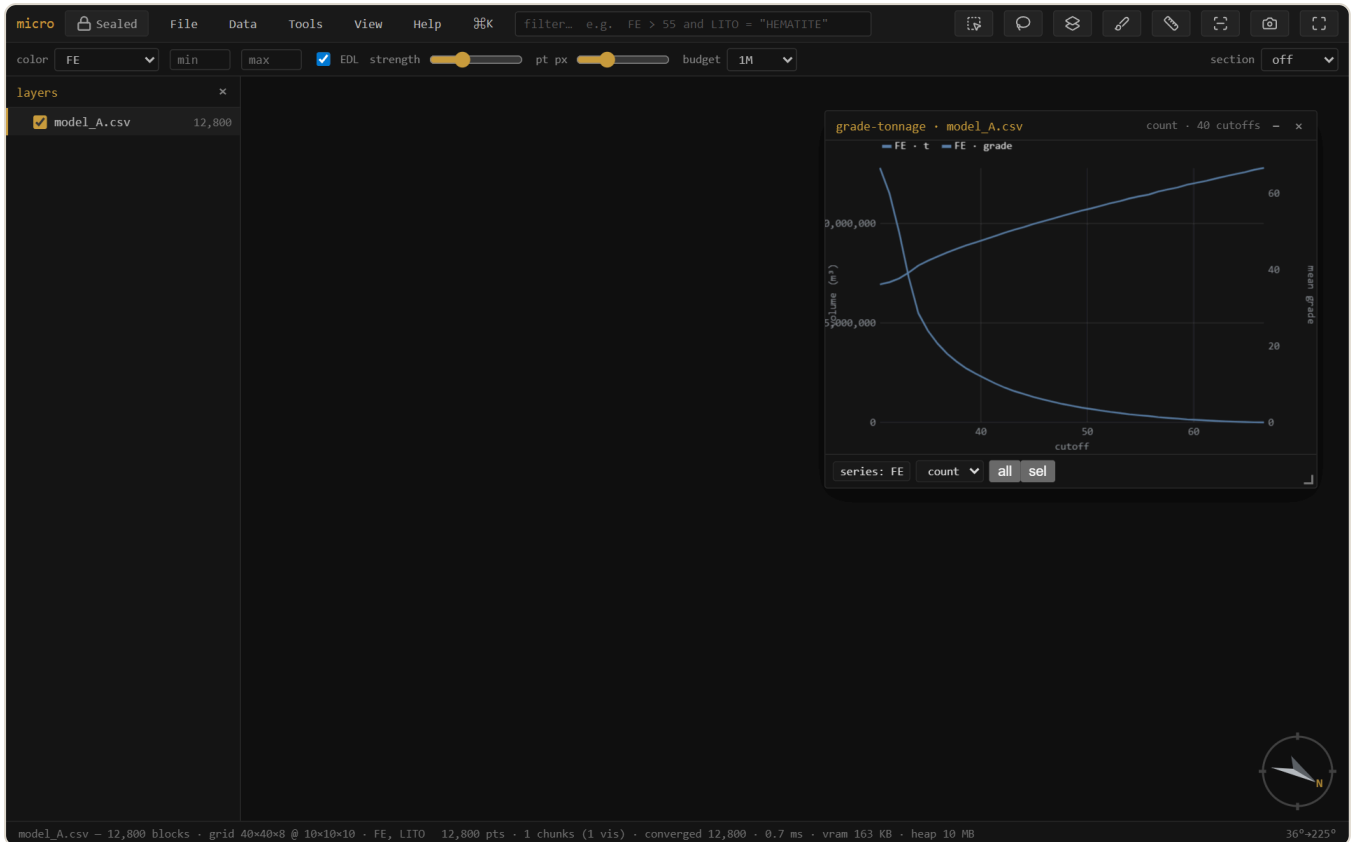


Reconciling two model versions. Global mean Δ is ≈ 0 , but the Δ map exposes a clear **conditional bias** — the new estimate runs low in the north (blue) and high in the south (red). A global check would have missed it entirely.

Tip — The Δ ramp auto-clips to the full range, which extreme blocks can widen. Tighten the **min/max** boxes (e.g. $\pm 2 \times$ the mean $|\Delta|$) to make the local pattern read, as above.

6 Grade-tonnage

Grade-tonnage... (right-click a model, or the palette) draws tonnage and mean grade above cutoff as a draggable plot. It is **tonnage-weighted** when you pick a density column ($\times$ block volume), else block count. Pick up to four grade columns; on a reconcile layer it auto-overlays reference vs new so you see how the grade-tonnage relationship shifted.

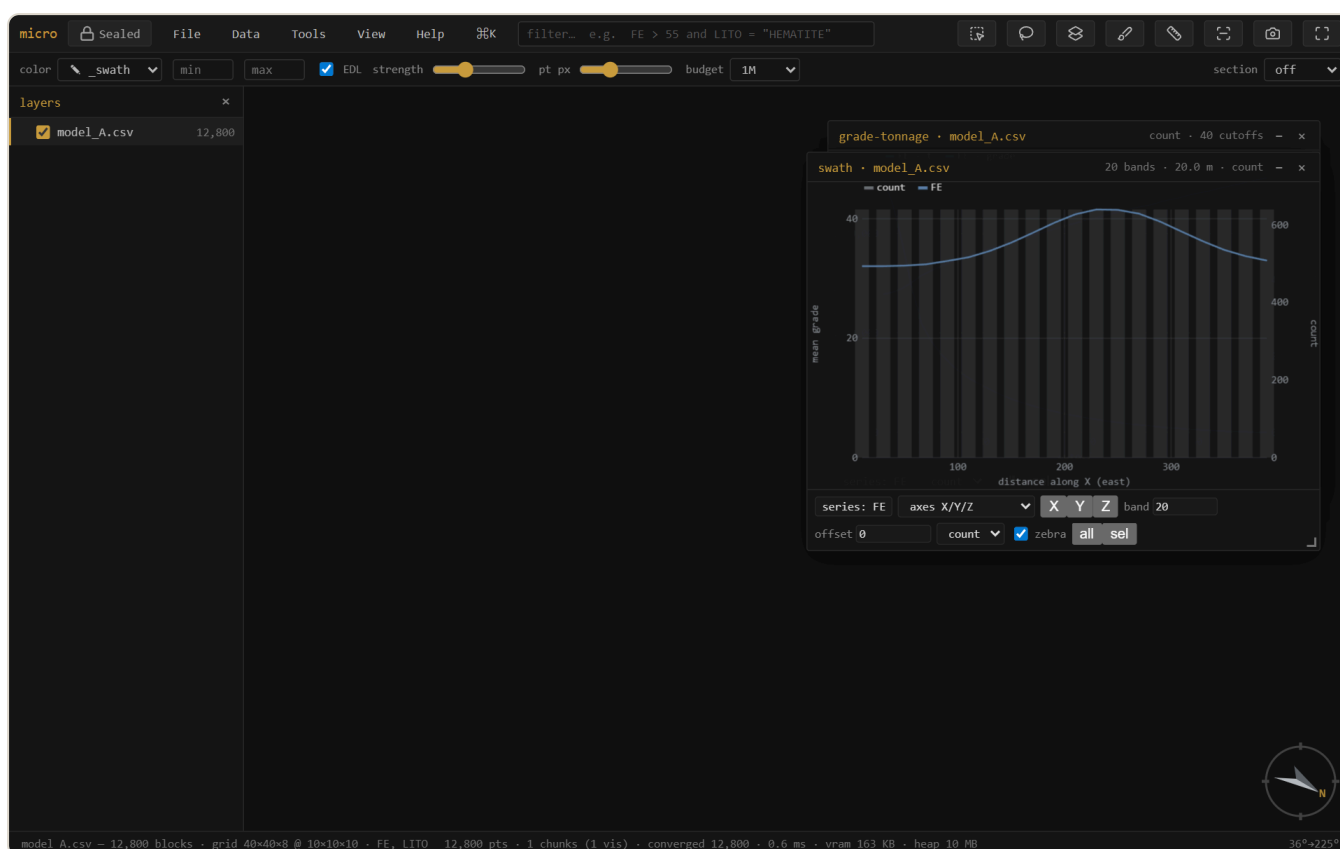


Grade-tonnage: tonnage (left axis) falls and mean grade (right axis) rises as the cutoff climbs — the recoverable-resource trade-off at a glance.

7 Swath / drift

Swath plot... shows mean grade per slab along a direction — the classic drift check for local bias. The direction can be the grid **axes** X/Y/Z, an oriented trio from **dip-direction / dip / rake** (across-structure · along-strike · down-dip), or a single **trend / plunge**. Band width and offset set the slabs; the count/tonnage bars sit behind the profile.

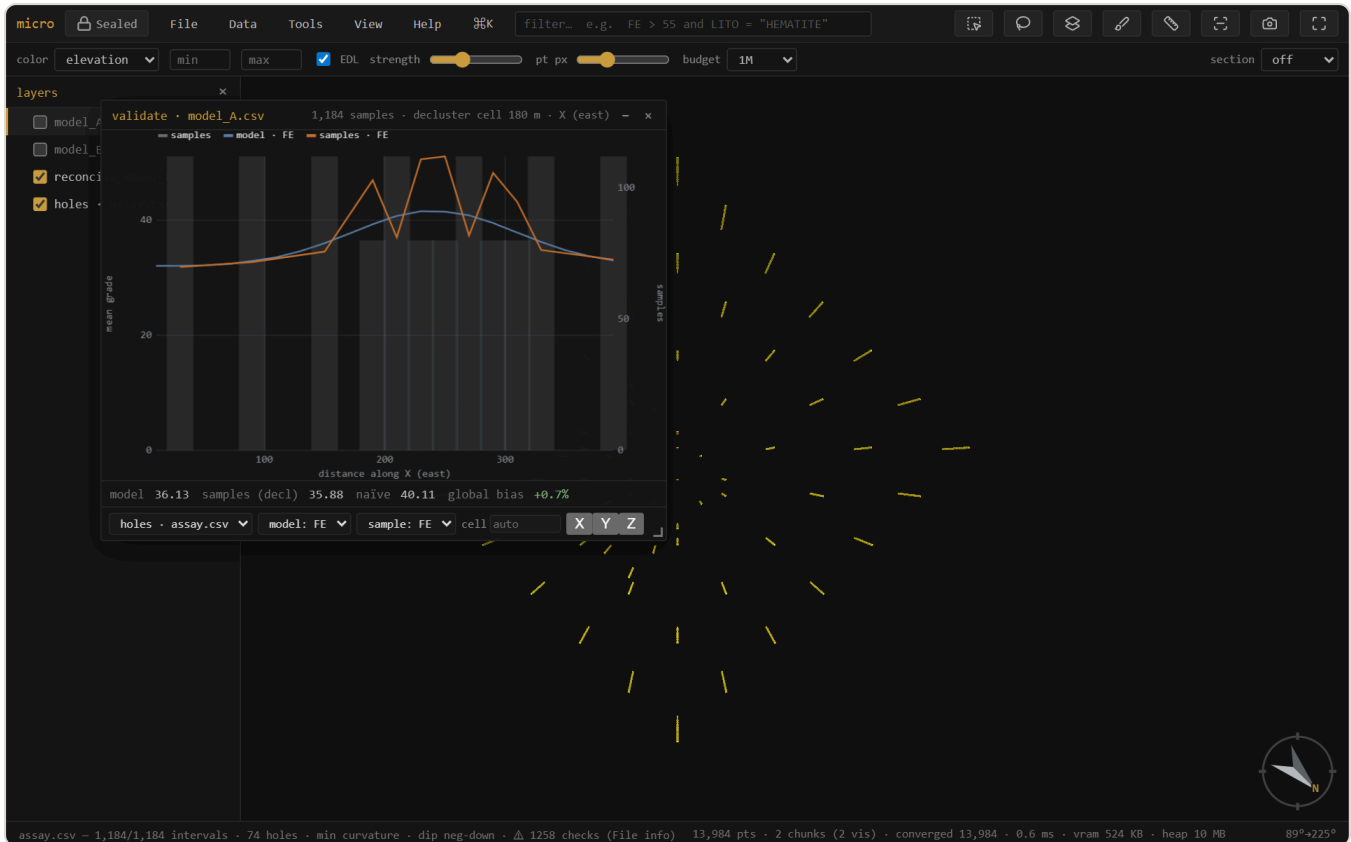
Zebra paints the alternate bands onto the 3D, so the slabs you see in space *are* the bins in the plot.



A swath along easting, with the bands zebra-striped onto the model. Because the plot and the stripes share one direction / width / offset, you read the profile against the deposit it came from.

8 Validate vs drillholes

Is the model consistent with the samples it was built from? **Validate vs drillholes...** takes a block model and a drillhole layer, **declusters** the sample grades (spatial sampling over-drills high grade, so a naïve sample mean is inflated), and reports the **global bias** (model mean vs declustered sample mean) plus a **model-vs-sample swath** along X/Y/Z.



Validation. The naïve sample mean (40.1) is inflated by clustered infill drilling; **declustered** it's 35.9, essentially matching the model (36.1) — a **+0.7% global bias**. The swath compares model (blue) and sample (orange) grade along easting, slab by slab.

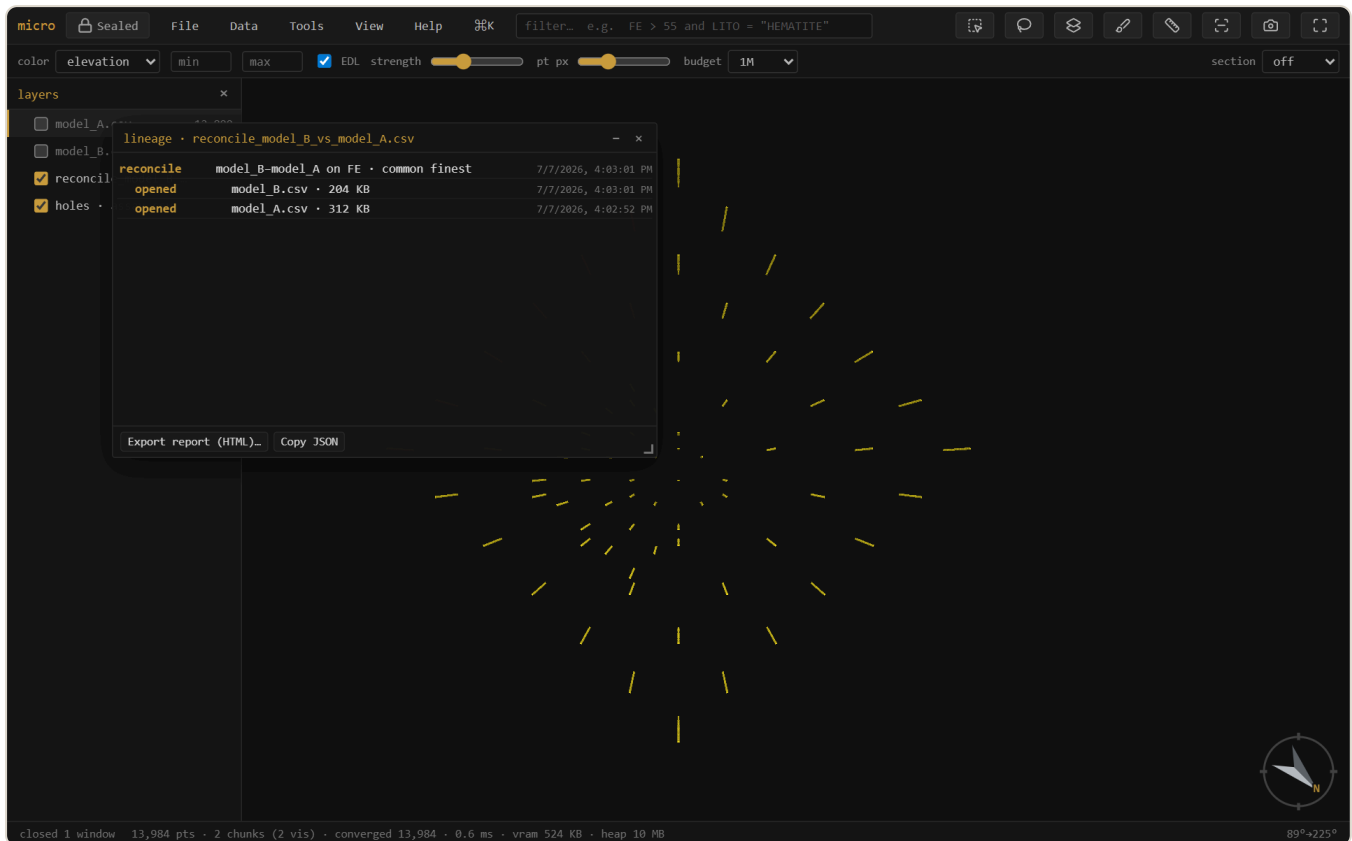
Note — Samples are point support, blocks are block support, so a *global* grade offset between them can be real (change of support), not an error. The swaths are what flag *local*, conditional bias.

9 Linked brushing

The grade-tonnage and swath windows carry an **all / selection** toggle. Set it to **selection**, then rubber-band a bench, a domain, or a suspect zone in the 3D — the plot recomputes for just those blocks, live, as you brush. The analysis becomes interactive regional exploration, not a whole-model summary.

10 Lineage & provenance

Every derived layer remembers *how it was made*. Opening a file is a leaf; a join, reconcile, or optimize wraps its source layers with the operation, its parameters, the build, and a timestamp — a tree. **Lineage...** shows that tree and exports a readable HTML **report** you can attach to a sign-off. Parquet exports carry the lineage embedded in the file, so the provenance travels *inside* the data.



The provenance tree for a reconciliation: the result wraps the two opened models. Export it as an HTML report, or read it back from any Parquet micro wrote.

11 Grids

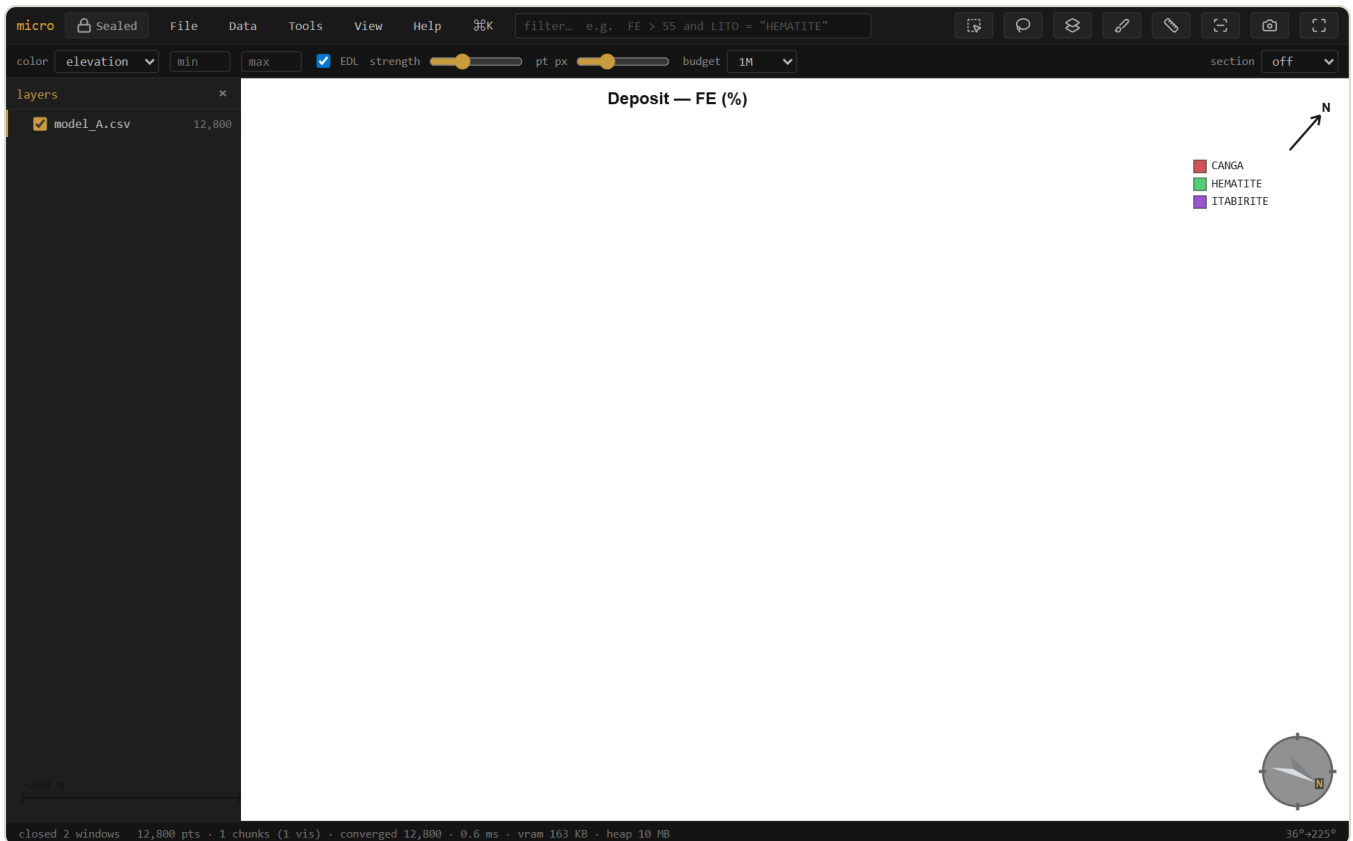
GeoTIFF / Surfer `.grd` / ESRI `.asc` open as heightfield layers (cells are pickable records; huge DEMs decimate for display while the full grid stays for evaluation). A mesh right-click → **Rasterize to grid...** renders it onto a DEM (the surface↔grid converter, with fill / extrapolation for gaps). **New...** → **grid / block model** creates empty grids from parameters, covering a layer's bounds, or — **cover compatible models** — on a lattice that covers and is compatible with several models at once (the reconcile target). Grid layers export as ESRI ASCII.

12 The Parquet project store

Optimize to spatial Parquet reorders a model Morton-sorted, so each row-group's footer becomes a tight spatial index and queries skip whole row-groups they can't match. This runs **automatically when you save a project** — Parquet is micro's project store. Painting stays editable through the reorder, and the whole thing round-trips. Parquet block models also open **streamed**, holding nothing decoded, with predicate and spatial push-down on the filter.

13 Figures & decorations

View → **Decorations & figure...** toggles a **scale bar**, **north arrow**, colour **legend**, and **title** onto the viewport — they show live and are captured verbatim by **Export figure**. Set a **background** (white or paper for print; the decorations flip to dark ink to stay readable). The exported PNG is what you see, ready for a report — and it pairs with the lineage report for a fully auditable figure.



A report-ready figure: white background, scale bar, north arrow, grade legend, and title — composited onto the export exactly as shown.

14 Projects & export

In a Chromium browser, **File** → **Open project folder** keeps your layers, styling, filters, camera, section, and saved views in a plain folder (`project.json` + the data files). `Ctrl+S` saves.

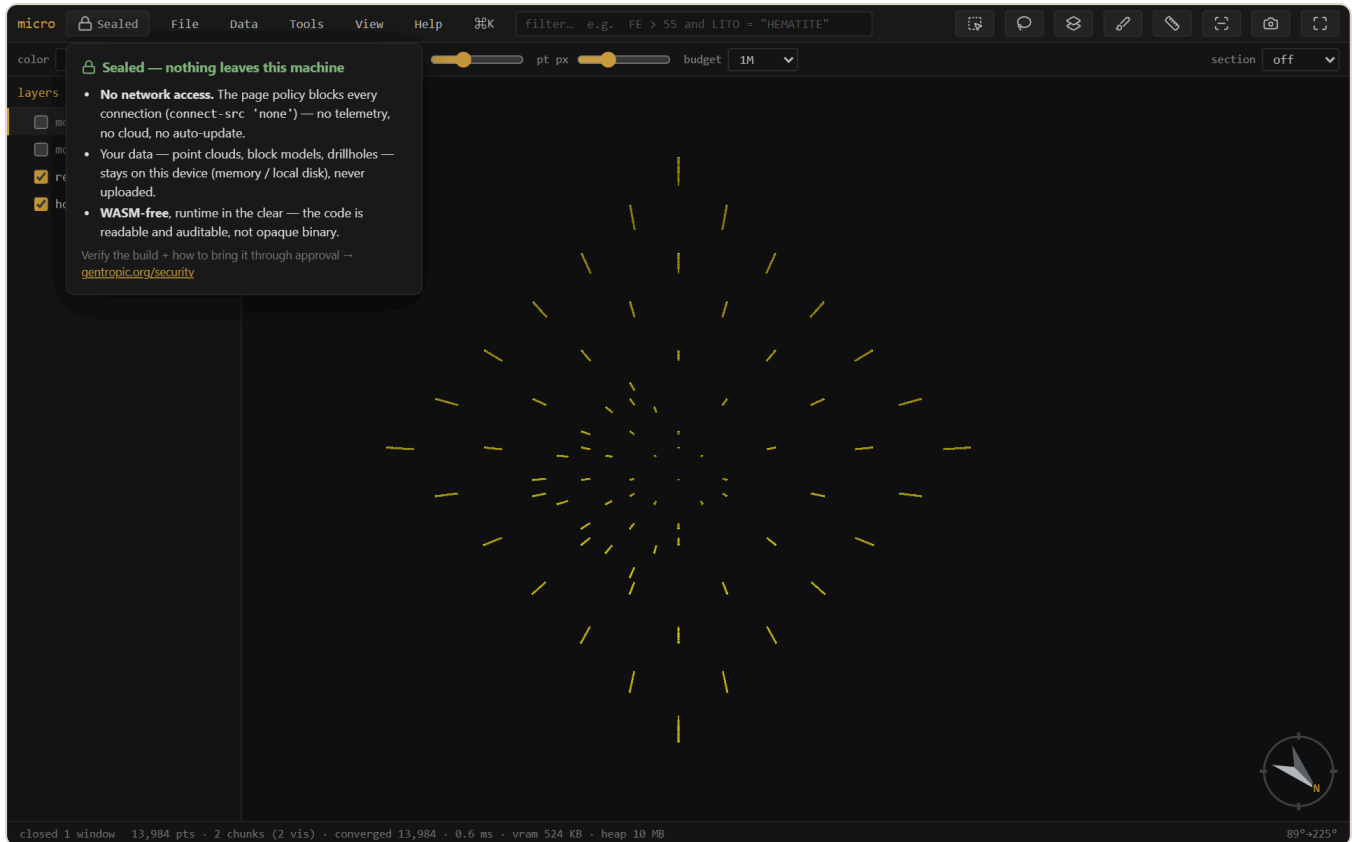
Export rows... writes any layer back out — scope **all** / **filter matches** / **selection**, every column including calculated and painted ones, as CSV or Parquet. Opening a `.dm` and exporting is the `.dm` → CSV converter.

15 Windows

The analysis, table, and paint windows are all draggable and pinned to their layer (open several at once). **View** → **Windows...** lists every open window — click to focus, or **Cascade** and **Close all** to tidy up.

16 Security: Sealed

micro is **Sealed**: it makes **no network requests** — a browser security policy (`connect-src 'none'`) blocks every connection, so there is no telemetry, no cloud, and no auto-update. Your data — point clouds, block models, drillholes — stays on your device. The build is **WASM-free** with the runtime in the clear, so the code is readable and auditable rather than opaque binary.



The **Sealed** badge (top-left) explains the posture and links to the verifier. For confidential resource data, this is the point: nothing leaves the machine.

Verify the build and read how to bring it through approval at gentropic.org/security.

17 Install & offline

micro runs at gentropic.org/micro — install it as an app from the browser, or download `micro.html` : one file, no installer, that you can keep, e-mail, drop on a shared drive, or run fully offline by double-clicking. There is nothing to set up and nothing that phones home. Works in current Chrome / Edge / Firefox; project folders and mounts need a Chromium browser.

A Reference

Keyboard

Key	Action	Key	Action
<code>⌘/Ctrl K</code>	command palette	<code>/</code>	focus the filter
<code>f</code>	fit the data	<code>p</code>	layers panel
<code>n s e w</code>	look level that way	<code>d</code> / <code>u</code>	plan / from below
<code>o</code>	ortho / perspective	<code>x</code>	section on/off
<code>k</code>	knife (drag a section)	<code>l</code>	face the section
<code>[</code> <code>]</code>	step section slabs	<code>m</code>	measure
<code>r</code> / <code>v</code>	select rectangle / lasso	<code>b</code>	paint a column
<code>t</code>	attribute table	<code>,</code> <code>.</code>	step the active layer
<code>Ctrl S</code>	save project	<code>Esc</code>	clear selection / cancel

The filter language

Comparisons `>` `>=` `<` `<=` `=` `!=`, boolean `and` / `or` / `not`, between a and b, in (a, b, c), over any column (text values in quotes). The same expressions define calculated columns. Autocomplete offers columns, functions, and category values — `Tab` / `Enter` accepts.

micro — part of the Auditable project, Geoscientific Chaos Union. Engine: `@gcu/condenser` ; declustering: `@gcu/gsjjs` . MIT.

Screenshots are generated from the app by an automated harness, so they track the current build. Docs for micro — see the in-app **Help** menu for the terse quick reference.